DEEP LEARNING LABORATORY

EXPERIMENT 13

SEQUENTIAL DATASET PREPARATION AND RNN PREDICTION

Aim

To prepare sequential time-series data and train a Simple Recurrent Neural Network (RNN) to predict the next value in a sine-wave sequence.

Output Summary

The sequence was converted into overlapping windows of 20 time steps. The resulting input tensor has 980 samples, 20 time steps, and one feature per step. The dataset was divided into 784 training samples and 196 testing samples. The SimpleRNN layer contains 288 parameters, and the Dense output layer contains 17 parameters, giving 305 trainable parameters in total.

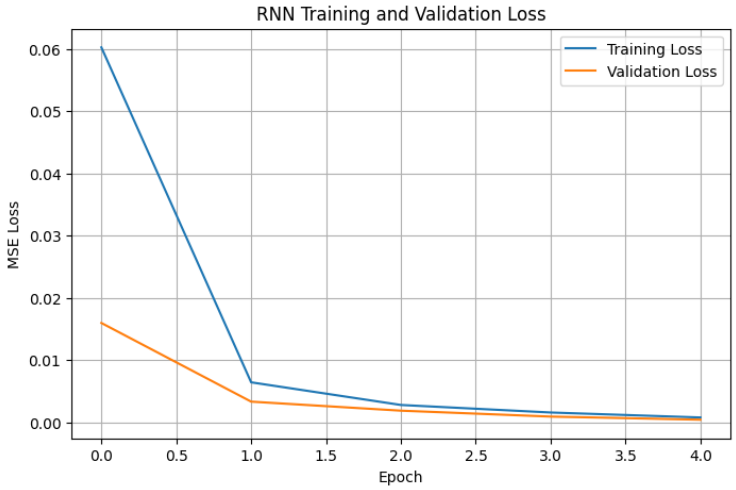
Recorded Results

Item	Recorded output
Input shape	(980, 20, 1)
Target shape	(980,)
Training / testing samples	784 / 196
SimpleRNN parameters	288
Dense parameters	17
Total trainable parameters	305
First five actual values	[-0.958, -0.925, -0.883, -0.831, -0.772]
First five predicted values	[-0.917, -0.872, -0.827, -0.785, -0.741]

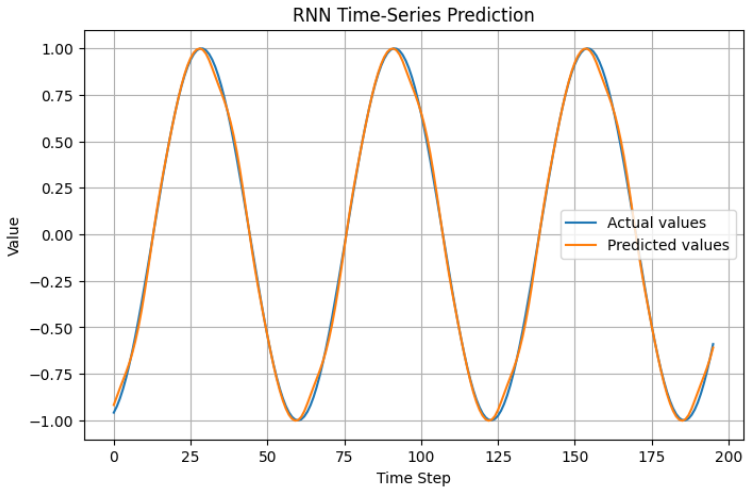
Google Colab Execution Screenshot



Training and Validation Loss



Actual vs. Predicted Time-Series Values



Result

The SimpleRNN model learned the sequential pattern in the generated sine-wave data. The prediction curve closely follows the actual values, while the training and validation loss decrease over the five epochs, indicating improved prediction performance during training.